

INSTRUCTOR'S MANUAL
AND
TEST BANK

to accompany

James W. Nybakken
Mark D. Bertness

MARINE BIOLOGY:
An Ecological Approach
Sixth Edition

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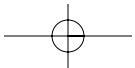
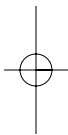
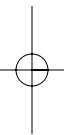
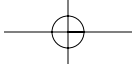


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PREFACE

As I stated in the preface to previous editions of this work, most instructors never bother to read the preface to a textbook, and I suspect that fewer still read or even use an Instructor's Manual and Test Bank (IM/TB). With this sort of opening, one might wonder why I have written this IM/TB, let alone a preface to the labor. It was done for a variety of reasons, some personal, some professional. Finally, I feel strongly that science education in the United States has taken a back seat to research and that this is inappropriate.

If, as Louis Pasteur said, only prepared minds make discoveries, science educators must ask who prepares those minds and how is that preparation best accomplished?

You should understand that my intent in writing this guide is in no way to save you the industry that must be employed in reading Nybakken's text for yourself. Nonetheless, it is appropriate for me to offer a different insight to the material and to provide sample questions that might help beginning instructors or even experienced ones develop an exam when time is precious; this is what I have attempted to do.

INSTRUCTOR'S MANUAL AND TEST BANK FORMAT

James Nybakken's sixth edition of *Marine Biology: An Ecological Approach* is an excellent text to introduce students to the discipline of marine biology. Naturally the material covered here reflects my own eclectic view of marine biology. Remember that this is a guide, nothing more; use it accordingly. This guide is divided into two parts. In the first section, I review each chapter (Chapter Outline and Significant Elements) and provide some Enrichment Exercises to support the material that is covered by the text; the second part is the Test Bank which is organized by separate chapters.

Chapter Outline and Significant Elements

Chapters are summarized and then reviewed section by section, touching on the most important aspects of the material being considered. Just as each chapter in Nybakken's text does not come in a standard size, I did not attempt to provide absolute equal coverage of every chapter.

The lists of terms or concepts (a.k.a. vocabulary, a horrible term in the minds of some students) is provided so that instructors can see at a glance an overview of the important concepts covered in the chapter. They also provide a basis for identification questions, should you choose to use that testing technique; I have not done so here. In fact, for the most part, I have avoided listing species names or even common names in these lists. Instructors should gauge for themselves the level of taxonomy germane to their course.

"'What's the use of their having names,' the Gnat said, 'if they won't answer to them?' 'No use to them,' said Alice; 'but its useful to the people that name them, I suppose.'"
(Lewis Carroll, *Through the Looking-Glass: and what Alice found there*. 1899)

Although listing of references is a job that the textbook does quite well, I have added some to provide ideas for discussion.

Enrichment Exercises

Enrichment Exercises that go beyond the general scope of the textbook are found at the end of each chapter. These have been reviewed from the last edition and updated as needed. I have continued the practice of not offering many Web sites, as one can never be sure that any URLs are up to date, or even still extant.

Test Bank Questions

The test bank component of this work now consists of more than 750 objective questions and more than 650 essay-discussion questions. With this diversity, instructors will be able to construct a variety of exams framed for their particular needs.

No attempt has been made to standardize the number of questions (of any type) per chapter. The types of questions I have chosen (matching, multiple choice, true-false, short answer, and essays) were not invented with any particular style of course in mind and they must not be taken as an end in themselves. I assume that no one will use these questions without making modifications. These questions are provided to stimulate instructors to invent their own questions. All of the questions provided in the test bank have been reviewed from the previous editions of the IG/TB. Where appropriate new ones have been invented, with the outcome of that labor being that about 30% of the questions are new or have been changed in some way. Essay questions supplied in the textbook are not duplicated in the Test Bank.

How you use these questions is up to you. Instructors with small class sizes may wish to use only essay questions. Institutions with substantially larger class sizes may reduce the number of essay questions, relying more on the objective style question. I do not fault instructors for using multiple choice questions, I use them myself when appropriate. Our time is limited and needs require that we be efficient with that commodity.

Please note that I have not attempted to randomize the position of the correct answers in the M/C questions (i.e., a, b, c . . .). Nearly all of them have letter “a” as the correct answer. I assume that instructors will go about the little job of randomization themselves. Matching questions are correctly aligned with their match. The result is that instructors should take care and review all of the objective questions should be adjusted before use. The Fill-in-the-Blank questions can, of course, be rewritten as M/C questions, but be careful, as there is some overlap of content between those two styles. Some of the essay questions I have invented may seem to you to be rather difficult, requiring additional reading and a good deal of class discussion. They do, and that is how I run my course. To encourage the students in this activity, I hand out all the essay questions at the beginning of the course and tell them that the exams will be based on those questions, with each chapter receiving 3–6 questions, depending on the length of the examination. New questions are added to the list and given to the students as they are invented.

Final Caveat and Acknowledgments

I have used all five editions of Nybakken’s text and, having access to the manuscript for the sixth, look forward to using it again. Although I worked from Jim Nybakken’s final copy, I take responsibility for any errors that occur in this IM/TB and welcome constructive criticism from instructors of marine biology.

I wish to acknowledge with much thanks James Nybakken who so many years ago suggested to the staff at HarperCollins that I write an instructor’s guide to his text. This small contribution to education is dedicated to all who love the ocean, even those of us who are landlocked.

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